Practice Questions

Assume that scientific studies provide you with the following information concerning the benefits and costs of sulfur dioxide emissions:

Benefits of abating (reducing) emissions: $MB = 500 - 20A$

Costs of abating emissions: $MC = 200 + 5A$

where A is the quantity abated in millions of tons and the benefits and costs are given in dollars per ton.

a. What is the socially efficient level of emissions abatement?

To find the socially efficient level of emissions abatement, set marginal benefit equal to marginal cost and solve for A:

$$500 - 20A = 200 + 5A$$

$$A = 12 \text{ million tons.}$$

b. What are the marginal benefit and marginal cost of abatement at the socially efficient level of abatement?

Substitute $A = 12$ into the marginal benefit and marginal cost functions to find the marginal benefit and cost:

$$MB = 500 - 20(12) = 260$$

$$MC = 200 + 5(12) = 260.$$

In a market for dry cleaning, the inverse market demand function is given by $P = 100 - Q$ and the (private) marginal cost of production for the aggregation of all dry-cleaning firms is given by $MC = 10 + Q$. Finally, the pollution generated by the dry-cleaning process creates external damages given by the marginal external cost curve $MEC = Q$.

a. Calculate the output and price of dry cleaning if it is produced under competitive conditions without regulation.

Set demand equal to supply to find the competitive equilibrium. To do this, set price equal to marginal cost (which is the industry supply function):

$$100 - Q = 10 + Q, \text{ so}$$

$$Q = 45, \text{ and } P = \$55.$$

b. Determine the socially efficient price and output of dry cleaning.

First, calculate the marginal social cost (MSC), which is equal to the marginal external cost plus the private marginal cost. Next, set MSC equal to the market demand function to solve for price and quantity. When all costs are included, the quantity produced will fall and the price will rise:

$$MSC = MC + MEC = (10 + Q) + Q = 10 + 2Q.$$

Setting $MSC = MSB$: $10 + 2Q = 100 - Q$, so

$$Q = 30, \text{ and } P = \$70.$$

TC = TFC + TVC

AC = AFC +AVC

A firm has a fixed production cost of $5000 and a constant marginal cost of production of $500 per unit produced.

a. What is the firm's total cost function? Average cost?

The variable cost of producing an additional unit, marginal cost, is constant at $500, so $VC = 500q$, and $AVC = \frac{VC}{q} = \frac{500q}{q} = 500$. Fixed cost is $5000 and therefore average fixed cost is $AFC = \frac{5000}{q}$. The total cost function is fixed cost plus variable cost or $TC = 5000 + 500q$. Average total cost is the sum of average variable cost and average fixed cost: $ATC = 500 + \frac{5000}{q}$.

b. If the firm wanted to minimize the average total cost, would it choose to be very large or very small? Explain.

The firm would choose to be very large because average total cost decreases as q is increased. As q becomes extremely large, ATC will equal approximately 500 because the average fixed cost becomes close to zero.

The short-run cost function of a company is given by the equation $TC = 200 + 55q$, where TC is the total cost and q is the total quantity of output, both measured in thousands.

a. What is the company's fixed cost?

When $q = 0$, $TC = 200$, so fixed cost is equal to 200 (or $200,000).

b. If the company produced 100,000 units of goods, what would be its average variable cost?

With 100,000 units, $q = 100$. Variable cost is $55q = (55)(100) = 5500$ (or $5,500,000). Average variable cost is $\frac{VC}{q} = \frac{5500}{100} = 55$ (or $55,000).

c. What would be its marginal cost of production?

With constant average variable cost, marginal cost is equal to average variable cost, which is 55 (or $55,000).

d. What would be its average fixed cost?

At $q = 100$, average fixed cost is $\frac{FC}{q} = \frac{200}{100} = 2$ (or $2000).

Economic cost = Accounting cost (explicit cost) + Implicit cost/Opportunity cost

A firm pays its accountant an annual retainer of $10,000. Is this an economic cost?

This is an explicit cost of purchasing the services of the accountant, so it is both an economic and an accounting cost. When the firm pays an annual retainer of $10,000, there is a monetary transaction. The accountant trades his or her time in return for money. An annual retainer is an explicit cost (as well as a fixed cost) and therefore an economic cost.